

Hengxu Guo(Rolzie)

hg1923@ic.ac.uk | www.rolzie.com | linkedin.com/in/hengxu-guo

SUMMARY

MEng Computing student at Imperial College London, expected First Class, with software engineering experience at Amazon Prime Video and HSBC across production web applications, backend services and testing infrastructure. Strong in TypeScript/React, Python and Java, with interests in product engineering and AI-assisted development.

EDUCATION

Imperial College London | *MEng in Computing (G401), Expected: First Class*

Sep 2023 – Jun 2027

Relevant modules: Software Engineering Design, Graphs & Algorithms, Computer Systems, Databases, Computing Practical (Kotlin, Java, C), Statistics.

EXPERIENCE

Software Engineer Intern | *Amazon (Prime Video), London*

Apr 2026 – Oct 2026

- Migrated Prime Video's Person Page (~50K daily global views) from a legacy widget-server architecture to a direct app-server SSR path, removing an intermediate service call from the critical rendering flow.
- Proposed and shipped an on-demand localisation feature for previously untranslated content, registering a new service operation, adding cache-backed frontend-backend requests and iteratively testing the flow across 33 non-English locales.
- Modernised the Person Page in TypeScript/React using reusable design-system components, applying AI-assisted workflows to explore unfamiliar frontend patterns and iterate faster on responsive Filmography UI.
- Improved frontend test reliability by migrating redesigned modules to Playwright, refactoring selectors and validating regressions across legacy and new rendering paths.

Software Engineer Intern | *HSBC*

Aug 2025 – Oct 2025

- Developed Java/Spring Boot backend testing tools for distributed financial-system integrations, including mock REST endpoints and authentication modules for secure API workflow simulation.
- Automated repetitive test-data preparation by building Python scripts to parse Excel datasets, extract required fields and populate testing procedures, reducing manual workload by 90%.
- Debugged integration issues across API, database and authentication layers, helping engineering teams in India and the UK validate service behaviour in a large-scale banking system.

LEADERSHIP & ACTIVITIES

First Commit — Bending Spoons | *Milan, Italy*

Nov 2026

- Selected as 1 of 30 student engineers to attend Bending Spoons' invite-only 5-day engineering retreat in Milan (Nov 2026), including a 3-day challenge to build a mobile or web app from scratch.

Optiver Trading Academy | *Imperial College London*

May 2025 – Jun 2025

- Applied market microstructure, arbitrage, and options-pricing concepts to design and back-test trading strategies.
- Built a delta-neutral strategy using arbitrage and volatility signals, ranking top among 30+ participants.

PROJECTS

Optimise My Coursework | *TypeScript / React*

Jan 2026 – Mar 2026

- Built a web extension for Imperial's Scientia course-selection platform, visualising expected workload across charts and calendars.
- Added configurable scheduling and personalised time-setting features to support better module planning.

Re@Imperial App | *React / Supabase*

May 2025 – Jun 2025

- Developed and iterated a React/Supabase web app from user research to live demo, refining the product through repeated user feedback to encourage reusable-container adoption.

WACC Compiler | *Scala*

Feb 2025 – Apr 2025

- Built a compiler for the WACC language, translating source code to ARM assembly via AST/TAC intermediate representations, with constant-folding optimisation via symbol propagation.

PintOS Operating System | *C*

Oct 2024 – Jan 2025

- Implemented core OS features (scheduling, system calls, process management, virtual memory) and debugged concurrency issues via synchronisation mechanisms.

SKILLS & INTERESTS

Languages: TypeScript, JavaScript, Python, Java, C, C++, Scala, SQL

Frameworks & Tools: React, Spring Boot, Playwright, Git, JUnit

Spoken Languages: Mandarin (Native), English (Fluent)

Interests: Swimming, Gliding, Badminton, Financial Markets